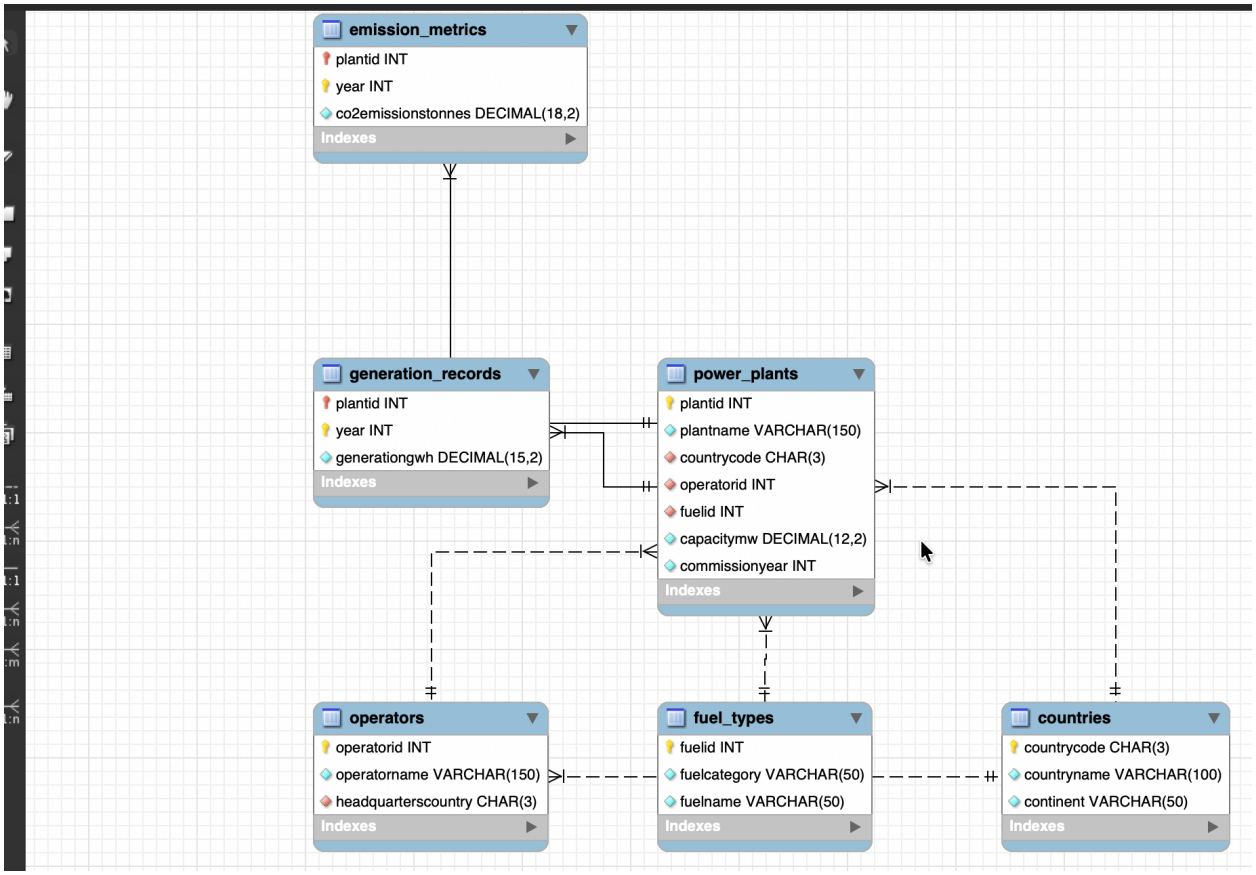
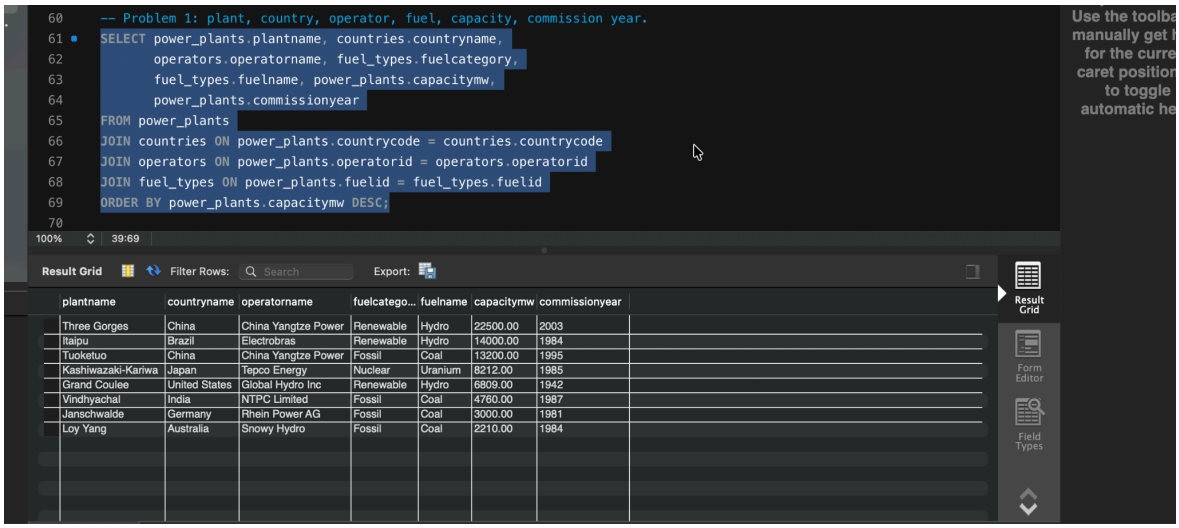
Luis Calvillo Assignment 3



Q1:



```
-- Problem 1: plant, country, operator, fuel, capacity, commission year.
SELECT power_plants.plantname, countries.countryname,
       operators.operatorname, fuel_types.fuelcategory,
       fuel_types.fuelname, power_plants.capacitymw,
       power_plants.commissionyear
FROM power_plants
JOIN countries ON power_plants.countrycode = countries.countrycode
JOIN operators ON power_plants.operatorid = operators.operatorid
JOIN fuel_types ON power_plants.fuelid = fuel_types.fuelid
ORDER BY power_plants.capacitymw DESC;
```

Result Grid

plantname	countryname	operatorname	fuelcatego...	fuelname	capacitymw	commissionyear
Three Gorges	China	China Yangtze Power	Renewable	Hydro	22500.00	2003
Itaipu	Brazil	Electrobras	Renewable	Hydro	14000.00	1984
Tuoketuo	China	China Yangtze Power	Fossil	Coal	13200.00	1995
Kashiwazaki-Kariwa	Japan	Tepco Energy	Nuclear	Uranium	8212.00	1985
Grand Coulee	United States	Global Hydro Inc	Renewable	Hydro	6809.00	1942
Vindhyachal	India	NTPC Limited	Fossil	Coal	4760.00	1987
Janschwalde	Germany	Rhein Power AG	Fossil	Coal	3000.00	1981
Loy Yang	Australia	Snowy Hydro	Fossil	Coal	2210.00	1984

Q2:

```sql
-- Problem 2: annual generation in 2024.
SELECT power_plants.plantname, power_plants.countrycode,
       generation_records.year, generation_records.generationgwh
FROM power_plants
JOIN generation_records ON power_plants.plantid = generation_records.plantid
WHERE generation_records.year = 2024
ORDER BY generation_records.generationgwh DESC;

-- Problem 3: generation and emissions for 2024.
SELECT power_plants.plantname, power_plants.countrycode,
```

Result Grid | Filter Rows: Search | Export:

plantname	countryco...	year	generationg...
Three Gorges	CHN	2024	98800.00
Itaipu	BRA	2024	89000.10
Tuoketuo	CHN	2024	71000.30
Vindhyachal	IND	2024	35600.80
Grand Coulee	USA	2024	21500.50
Janschwalde	DEU	2024	18400.20
Loy Yang	AUS	2024	14200.40
Kashiwazaki-Kariwa	JPN	2024	0.00

Q3:

```sql
SELECT power_plants.plantname, power_plants.countrycode,
       generation_records.year, generation_records.generationgwh,
       emission_metrics.co2emissionstonnes
FROM power_plants
JOIN generation_records ON power_plants.plantid = generation_records.plantid
JOIN emission_metrics ON power_plants.plantid = emission_metrics.plantid
    AND generation_records.year = emission_metrics.year
WHERE generation_records.year = 2024
ORDER BY emission_metrics.co2emissionstonnes ASC;

-- Problem 4: operator totals across all years (CTE, no table aliases).
```

Result Grid | Filter Rows: Search | Export:

plantname	countryco...	year	generationg...	co2emissionstonn...
Grand Coulee	USA	2024	21500.50	0.00
Itaipu	BRA	2024	89000.10	0.00
Kashiwazaki-Kariwa	JPN	2024	0.00	0.00
Three Gorges	CHN	2024	98800.00	0.00
Loy Yang	AUS	2024	14200.40	1950000.00
Janschwalde	DEU	2024	18400.20	2450000.00
Vindhyachal	IND	2024	35600.80	3100000.00
Tuoketuo	CHN	2024	71000.30	5200000.00

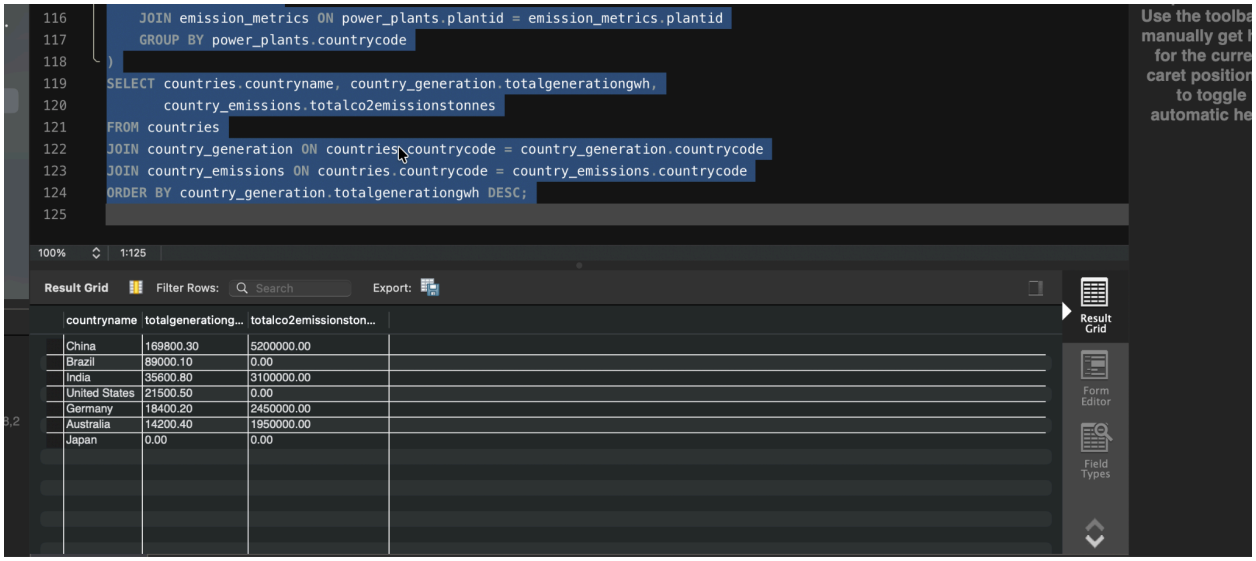
Q4:

```sql
WITH operator_generation AS (
    SELECT power_plants.operatorid,
           SUM(generation_records.generationgwh) AS totalgenerationgwh
    FROM power_plants
    JOIN generation_records ON power_plants.plantid = generation_records.plantid
    GROUP BY power_plants.operatorid
)
SELECT operators.operatorname, operators.headquarterscountry,
       operator_generation.totalgenerationgwh
FROM operators
JOIN operator_generation ON operators.operatorid = operator_generation.operatorid
```

Result Grid | Filter Rows: Search | Export:

operatorname	headquarterscoun...	totalgeneratiog...
China Yangtze Power	CHN	169800.30
Electrobras	BRA	89000.10
NTPC Limited	IND	35600.80
Global Hydro Inc	USA	21500.50
Rhein Power AG	DEU	18400.20
Snowy Hydro	AUS	14200.40
Tepco Energy	JPN	0.00

Q5:


JOIN emission_metrics ON power_plants.plantid = emission_metrics.plantid
GROUP BY power_plants.countrycode
)
SELECT countries.countryname, country_generation.totalgenerationgwh,
country_emissions.totalco2emissionstonnes
FROM countries
JOIN country_generation ON countries.countrycode = country_generation.countrycode
JOIN country_emissions ON countries.countrycode = country_emissions.countrycode
ORDER BY country_generation.totalgenerationgwh DESC;
100%
1:125
Result Grid
Filter Rows:
Search
Export:
countryname
totalgeneration...
totalco2emissionston...
China 169800.30 5200000.00
Brazil 89000.10 0.00
India 35600.80 3100000.00
United States 21500.50 0.00
Germany 18400.20 2450000.00
Australia 14200.40 1950000.00
Japan 0.00 0.00
Result Grid
Form Editor
Field Types